

Priyansh Gupta

Senior Agentic AI Engineer | GenAI Lead | MLOps | Multi-Agent Systems

Senior Agentic AI & GenAI Engineer with 5+ years of experience delivering enterprise AI solutions on Azure, AWS. Expertise in RAG, multi-agent systems, MCP integrations, and Azure OpenAI. Proven success building production-ready GenAI platforms for Fortune 500 clients.

@ priyansh.9071@gmail.com

in priyansh19

🔗 priyansh19

📞 +971 506 479 919

WORK EXPERIENCE

Senior DevOps & AI Platform Engineer

🏢 Applied Information Services (AIS) 📅 Aug 2024 – Mar 2026 📍 Hyderabad, India

Keywords: Azure AI Foundry, Azure OpenAI, RAG, MCP, Azure AI Search, DevOps

- Deployed multimodal LLM models on Azure AI Foundry, configuring model endpoints, inference auto-scaling, and Azure AI Content Safety guardrails for production AI workloads.
- Architected enterprise RAG pipelines using document ingestion, chunking, and vector plus semantic hybrid indexing in Azure AI Search, improving grounded responses for Fortune 500 workloads.
- Delivered MCP integration for Azure Boards, enabling AI assistants to reference live tickets, sprint data, and work items for enterprise agentic tool-use.
- Led GenAI platform modernisation for FMGlobal, supporting zero-downtime migration of critical insurance workloads to secure Azure AI infrastructure.
- Standardised CI/CD governance across 20+ microservices and deployed JFrog Artifactory with Docker image promotion gates for container supply-chain auditability.

Senior GenAI / MLOps / Agentic AI Consultant

🏢 Freelance Consulting (Upwork) 📅 Aug 2022 – Present 📍 Remote – Global Clients

Keywords: LangChain, Pinecone, ChromaDB, Azure OpenAI, Claude, GPT-4o, MLOps

- Architected end-to-end RAG pipelines using LangChain with Pinecone and ChromaDB for semantic retrieval, top-k reranking, and grounded responses for production SaaS clients.
- Delivered production agentic AI solutions embedding Azure OpenAI, Claude, and GPT-4o into SaaS workflows with tool-use, function-calling, and autonomous decision-making.
- Deployed MLOps infrastructure on Azure ML and AWS SageMaker, orchestrating agentic pipelines with n8n and Python for model monitoring and drift detection.

DevOps Engineer

🏢 Applied Information Services (AIS) 📅 Oct 2022 – Aug 2024 📍 Hyderabad, India

Keywords: AKS, Helm, Kyverno, Istio, FluxCD, Azure Key Vault

- Engineered Helm Charts for AKS onboarding with rollback-enabled pipelines and implemented Kyverno admission control for security baselines, resource quotas, and image signing.
- Configured Istio service mesh with mTLS zero-trust peer authentication, FluxCD GitOps state management, and canary plus A/B traffic routing.
- Managed secrets via Kubernetes Secrets Store CSI Driver and Azure Key Vault, enforcing least-privilege access across AKS clusters.

Software Engineer

🏢 Applied Information Services (AIS) 📅 May 2021 – Oct 2022 📍 Hyderabad, India

Keywords: Terraform, Bicep, Azure DevOps, YAML, AKS, Key Vault

- Built reusable Terraform and Bicep modules for Azure VNets, Key Vaults, AKS, and Application Gateways.
- Authored Azure DevOps YAML CI/CD templates for build-test-deploy workflows across cloud-native services.

CERTIFICATIONS

AI-102 Azure AI Engineer Associate · AZ-400 DevOps Engineer Expert · CKA Certified Kubernetes Administrator · CKAD Certified Kubernetes Application Developer

AZ-700 Azure Network Engineer · AZ-104 Azure Administrator · AI-900 Azure AI Fundamentals · AZ-900 Azure Fundamentals

EDUCATION

M.Sc. Artificial Intelligence

🏛️ Heriot-Watt University 📅 In Progress 📍 Dubai, UAE

- Focus on agentic AI systems and GenAI applications.

B.Tech. Computer Science

🏛️ University of Petroleum and Energy Studies 📅 2017 – 2021

- Specialized in cloud computing and AI/ML coursework.

KEY PROJECTS

Mach2 – Multi-Agent ReAct Harness with Observability & LLM-Response Evaluation

[Github Link](#)

LLM-as-a-Judge (Grok) · Multi-Layer Critic Pipeline · Langfuse Observability · LangGraph ReAct · Electron + React

- Architecting Mach2 (successor to Mach1) as a multi-agent ReAct harness instrumented end-to-end with self-hosted Langfuse – tracing every thought, tool call, and LLM span, with an LLM-as-a-Judge (Grok) scoring each response on hallucination and quality to surface live evaluation and performance metrics.
- Designed a multi-layer Critic pipeline that evaluates every LLM response before it reaches the user – Fact, Gap, Logic, Efficiency, and Sign-off critics emit a severity-tagged critique ledger (blocker/major/minor) that gates synthesis and triggers targeted re-research, maximising answer quality per token.
- Built a nested Research Agent that dynamically fans out scoped scouting sub-agents (agent-as-tool), backed by shared ChromaDB memory and an Electron + React desktop UI with a live Thinking & Critique panel streaming agent reasoning and critic verdicts over SSE.

Mach1 – Local-First Multi-Tool AI Agent with Triage Routing

[Github Link](#)

LangChain · LangGraph (StateGraph) · Ollama (Gemma) · ChromaDB · FastAPI · React + TypeScript · RAG Memory

- Built Mach1, a local-first AI agent on LangChain and LangGraph (StateGraph) running open-source LLMs through Ollama, with a confidence-scored triage router that answers 65%+ of requests locally and escalates only hard prompts to Claude (via MCP) – fast, private, and zero-cost.
- Engineered a full multi-tool agent loop (file ops, shell, Python REPL, DuckDuckGo, Wikipedia, Stack Overflow, URL browsing) and a layered memory system – episodic and semantic vector stores in ChromaDB with Top-k RAG recall and a periodic summarizer agent – served over a FastAPI backend.
- Designed a three-mode React + TypeScript (Vite) UI to drive the harness: multi-session sidebar, per-message tool badges, live confidence and routing tags, memory indicators, and a live-adjustable threshold slider.

Ornith 1.0 – Local Coding-Agent Harness for the Self-Scaffolding Ornith Model

[Github Link](#)

Self-Scaffolding ornith:9b · Pluggable Python Harnesses · Native Tool-Calls · ReAct + Critic Ledger · Next.js + Electron

- Built Ornith 1.0, a desktop coding-agent harness (Next.js + Electron, React + TypeScript) for the self-scaffolding ornith:9b model on local Ollama, where each session binds its own pluggable Python agent harness that plans tasks and streams native tool-calls (file edits, reads, shell runs).
- Built an Inspector / Thinking / Critiques workspace surfacing the agent's ReAct reasoning and an L2–L7 critic ledger per response, with session/folder organisation, a harness registry (load/unload), and a full Playwright end-to-end test suite.

BlastR – Graph-Aware RAG Security Analysis for Cross-Layer DevSecOps (M.Sc. Dissertation)

[Github Link](#)

LangGraph · LoRA/QLoRA Fine-Tuning · HuggingFace Transformers · FAISS RAG · NetworkX Attack Graphs · FastAPI · React

- Built BlastR (M.Sc. dissertation) that parses Terraform/IaC into NetworkX attack graphs and propagates "blast radius" across cloud layers, pairing a LangGraph reasoning agent with FAISS + Sentence-Transformers RAG to explain and prioritise security misconfigurations.
- Fine-tuned a domain SLM (Qwen2.5-Coder-1.5B) with LoRA/QLoRA via HuggingFace Transformers + PEFT on Terraform labelled with Checkov and Trivy, benchmarked against a zero-shot baseline and tracked in LangSmith; shipped a Typer CLI, FastAPI service, and React + TypeScript dashboard.

CrewSpace – Open-Source Multi-Agent AI Workforce Platform

[Github Link](#)

Hierarchical orchestration · Persistent memory graph · 3D Workspace · Agent Collaboration

- Architected and built CrewSpace from scratch, an AI workforce platform where a CEO agent autonomously decomposes user goals into domain-specific subtasks, delegates work across specialized AI agents, dynamically proposes new hires when required, and coordinates execution through multi-agent communication and approval workflows.
- Developed persistent per-agent memory, agent-to-agent and group chat, shared workspace and file permissions management, integrated terminal operations, real-time task boards, founder dashboards, and an interactive 3D office environment enabling users to monitor, collaborate with, and manage long-running AI organizations at scale.

TECHNICAL SKILLS

Agentic AI & Frameworks

LangChain · LangGraph (StateGraph) · CrewAI · Pydantic-AI · AutoGen · LlamaIndex · MCP · ReAct · Multi-Agent Orchestration · Tool-use · HITL · Guardrails · Chain-of-Thought

LLM Fine-Tuning & ML

LoRA / QLoRA · HuggingFace Transformers · PEFT · PyTorch · Datasets · Accelerate · Quantization · Instruction Tuning · Qwen2.5-Coder · Sentence-Transformers

LLM Eval & Observability

Langfuse · LangSmith · LLM-as-Judge & Critic Pipelines · Tracing · Confidence Scoring · Zero-shot Eval · Triage Routing · Prompt-Injection Defense

Models & Inference

Ollama · LM Studio · Gemma · Qwen · Azure OpenAI (GPT-4o) · Claude · Grok

RAG & Vector Stores

ChromaDB · FAISS · Pinecone · Qdrant · Azure AI Search · HuggingFace & nomic Embeddings · Episodic + Semantic Memory · Top-k Reranking

Azure AI Platform

Azure AI Foundry · Azure ML · Prompt Flow · Azure AI Content Safety · Copilot Studio

Backend & UI

Python · FastAPI · Typer · Pydantic v2 · SQLAlchemy · NetworkX · Node.js · TypeScript · React · Next.js · Vite · Electron · REST · Playwright

Cloud, IaC & DevOps

Azure · AWS · GCP · Terraform · Bicep · Azure DevOps · GitHub Actions · ArgoCD · FluxCD

Containers & Security

Kubernetes (AKS, EKS) · Docker · Helm · Istio · Kyverno · OPA · Checkov · Trivy · RBAC · mTLS Zero-trust · Azure Key Vault · Prometheus · Grafana